

Monocular Visual-Inertial Odometry on the TUM-VI Dataset

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Abstract—This paper presents the implementation and evaluation of a classical monocular visual odometry (VO) pipeline with lightweight inertial fusion using the TUM-VI benchmark dataset. The proposed system estimates camera motion from sequential monocular images using ORB feature extraction, feature matching, and geometric verification via the Essential Matrix with RANSAC. A simplified IMU-based motion model is incorporated to provide short-term pose smoothing between visual updates. The system is evaluated on the Room2, Corridor3, and Outdoor5 sequences of the TUM-VI dataset. Experimental results show that the addition of inertial measurements provides only marginal improvement over pure visual odometry, indicating that the current fusion strategy acts mainly as a smoothing component rather than a full drift-correction mechanism.

Index Terms—Visual Odometry, Visual-Inertial Odometry, Monocular Vision, ORB Features, IMU Sensor Fusion, TUM-VI Benchmark

I. INTRODUCTION

Real-time metric state estimation is a fundamental problem in autonomous robotics, enabling applications such as navigation, mapping, and motion tracking. Visual Odometry (VO) estimates ego-motion from sequential camera images and serves as a core component in modern Simultaneous Localization and Mapping (SLAM) systems.

Despite its effectiveness, monocular VO suffers from inherent limitations, particularly scale ambiguity and drift accumulation over time. These issues arise due to the inability of a single camera to directly observe depth, making long-term trajectory estimation challenging in unconstrained environments.

To address these limitations, Visual-Inertial Odometry (VIO) combines visual information with inertial measurements from an Inertial Measurement Unit (IMU). The IMU provides high-frequency motion information that improves short-term motion consistency and helps stabilize pose estimation during rapid movements or temporary visual degradation.

This project focuses on the implementation of a classical geometry-based monocular VO pipeline enhanced with lightweight IMU-based motion smoothing. The system is designed to remain computationally efficient while maintaining stable trajectory estimation without relying on complex optimization frameworks or deep learning methods.

The visual frontend is based on ORB feature extraction, feature matching, and motion estimation using the Essential

matrix with RANSAC outlier rejection. The IMU data is integrated in a simplified manner to provide short-term motion stabilization between frames.

The system is evaluated on the Room2, Corridor3, and Outdoor5 sequences of the TUM-VI benchmark dataset. Experimental results show that IMU integration provides only marginal improvement over pure visual odometry, indicating that the system is primarily vision-driven.

- Implementation of a monocular visual odometry frontend based on ORB feature extraction and geometric verification using the Essential matrix with RANSAC.
- Integration of IMU measurements for lightweight motion smoothing between visual updates.
- Evaluation on Room2, Corridor3, and Outdoor5 sequences from the TUM-VI benchmark dataset.

II. RELATED WORK

Visual localization has evolved significantly over the past two decades, progressing from geometric monocular vision systems to modern multi-sensor fusion approaches. Early visual odometry methods relied primarily on feature-based pipelines, where camera motion is estimated from frame-to-frame correspondences using epipolar geometry and outlier rejection techniques such as RANSAC.

More recent approaches extend these monocular frameworks by incorporating inertial measurements from IMU sensors, forming Visual-Inertial Odometry (VIO) systems. These methods improve robustness in scenarios with rapid motion or temporary visual degradation by providing high-frequency motion priors.

Classical optimization-based pipelines, such as sliding-window bundle adjustment and factor graph formulations, further improve accuracy by jointly minimizing visual reprojection and inertial residuals. However, these methods often introduce increased computational complexity.

In contrast, lightweight implementations such as the one presented in this work prioritize computational efficiency by combining monocular visual odometry with simplified inertial fusion, without employing full optimization-based backend architectures.

A. Monocular Feature Extraction and Tracking (Task 1)

Early visual odometry (VO) frameworks relied heavily on dense optical flow tracking variants or classical corner detectors, such as Harris corners or Shi-Tomasi features, to isolate and follow salient landmarks across sequential image frames [5]. While computationally lightweight, these initial methodologies demonstrated extreme vulnerability to sudden exposure transitions and entirely lacked scale invariance. The introduction of scale-invariant descriptor pipelines, most notably the Scale-Invariant Feature Transform (SIFT) and Speeded Up Robust Features (SURF), drastically improved tracking reliability under large baseline changes and perspective alterations. However, the high computational footprint associated with floating-point descriptor calculation restricted real-time deployment on standard embedded processing hardware.

To satisfy the stringent processing budgets of real-time robotics, contemporary geometric state estimators heavily utilize binary descriptors. Rublee et al. formulated Oriented FAST and Rotated BRIEF (ORB) as a computationally efficient, rotation-invariant, and noise-resilient alternative to SIFT [3]. In classical monocular configurations, feature continuity is preserved across sequential frames either via sparse pyramidal Lucas-Kanade optical flow tracking or explicit descriptor matching evaluated through Hamming distance parameters. To filter out the vast amounts of erroneous data associations generated during rapid spatial movements, modern pipelines implement Lowe’s ratio test alongside rigorous bidirectional cross-checking verification. This checks for consistency from the primary to the secondary frame and vice versa, maintaining a clean visual tracking database prior to coordinate resolution.

B. Epipolar Geometry and Relative Pose Estimation (Task 2)

Recovering relative spatial rotation ($\mathbf{R}$) and translation ($\mathbf{t}$) transformations between uncalibrated or calibrated subsequent views is dictated by foundational epipolar geometric relations. Longuet-Higgins pioneered the classical 8-point algorithm to resolve the relative camera displacement matrix, defined as the Essential Matrix ($\mathbf{E}$), for intrinsically calibrated systems where raw tracking coordinates are normalized via the camera intrinsic parameter matrix $\mathbf{K}$ [?]. For raw uncalibrated camera configurations, the Fundamental Matrix ($\mathbf{F}$) directly maps matching pixel coordinates across displacement lines without requiring structural normalization.

Because visual feature trackers are inherently corrupted by noise and false correspondences, direct algebraic solvers collapse if applied to raw input streams. To robustly circumvent this vulnerability, Fischler and Bolles established the Random Sample Consensus (RANSAC) iteration framework [8]. This algorithm isolates the maximum consensus set of true feature inliers that satisfy the algebraic epipolar constraint equation ($\mathbf{x}^T \mathbf{E} \mathbf{x} = 0$). Following RANSAC purification, Singular Value Decomposition (SVD) factors the Essential Matrix into four distinct kinematic hypotheses for relative pose $[\mathbf{R}|\mathbf{t}]$. The historic physical ambiguity is resolved via an explicit chirality test, which triangulates a sparse point subset and enforces a

positive depth condition ($Z > 0$) along the optical axes of both focal points.

C. Scale Ambiguity and Monocular Limitations

Monocular visual odometry is inherently unable to recover absolute metric scale from a single moving camera without additional sensors or prior information. As a result, the estimated trajectory is only defined up to an unknown scale factor.

In scenarios involving low parallax motion, such as pure forward translation or movement through texture-sparse environments (e.g., long corridors), the lack of sufficient geometric constraints leads to increased uncertainty in depth estimation. This often results in accumulated drift over time, especially in the absence of loop closure or global optimization.

To mitigate short-term instability, the system relies on feature-based motion estimation using epipolar geometry and RANSAC outlier rejection. However, without true metric depth observability, the recovery of absolute scale remains fundamentally unobservable in a monocular setting.

III. DATASET AND EXPERIMENTAL SETUP

The efficacy of the proposed tightly-coupled visual-inertial odometry pipeline is evaluated using the TUM-VI benchmark [8]. This dataset is selected for its high-fidelity sensor synchronization, challenging dynamic environments, and ground truth availability, which are essential for validating the metric accuracy of optimization-based state estimation.

A. Sensor Configuration and Data Acquisition

The TUM-VI dataset provides synchronized monocular image sequences and inertial measurements acquired using a calibrated sensor rig.

- **Monocular Camera:** A global-shutter camera providing grayscale images at a resolution of 512×512 pixels, captured at approximately 20 Hz. The visual frontend relies solely on this monocular stream for motion estimation using feature-based methods.
- **Inertial Measurement Unit (IMU):** A high-frequency MEMS IMU providing synchronized accelerometer and gyroscope measurements at 200 Hz. These measurements are used to improve short-term motion consistency between consecutive visual frames through lightweight fusion.

B. Evaluation Sequences

We evaluate the pipeline on three representative sequences to demonstrate performance across varying structural geometries:

- 1) **Room2:** A high-speed, indoor motion capture environment that provides dense, ground-truth-validated trajectories. This sequence evaluates the frontend’s ability to maintain localized metric precision during rapid rotation and translation.
- 2) **Corridor3:** A challenging low-texture indoor environment characterized by long, repetitive hallway structures and limited visual features. This sequence is particularly

difficult for monocular visual odometry due to reduced parallax and weak geometric constraints, which can lead to increased drift over time. It is used to evaluate the robustness of feature tracking and motion estimation under degenerate visual conditions.

- 3) **Outdoor5:** A large-scale outdoor sequence featuring varying motion patterns and environmental conditions, including illumination changes and wider spatial movement. This sequence is used to evaluate the long-term stability of the monocular visual odometry system under real-world conditions. Due to increased motion complexity and environmental variability, it provides a challenging scenario for maintaining consistent feature tracking and reducing cumulative drift over time.

C. Performance Metrics

The system's tracking precision is quantified using the Absolute Trajectory Error (ATE) and Relative Pose Error (RPE). These metrics measure the alignment between the estimated trajectory and ground truth data using the Umeyama SVD-based alignment algorithm. By focusing on these metrics, we assess the pipeline's robustness against bias accumulation, scale instability, and geometric degradation across different environmental challenges.

The implementation was developed in Python using OpenCV, NumPy, and SciPy.

IV. SYSTEM PIPELINE

The system pipeline is architected as a sequential flow of data, from sensor ingestion to state estimation. The architecture is illustrated in Figure 1, representing the modular separation between the visual frontend, the inertial integration, and the optimization backend.

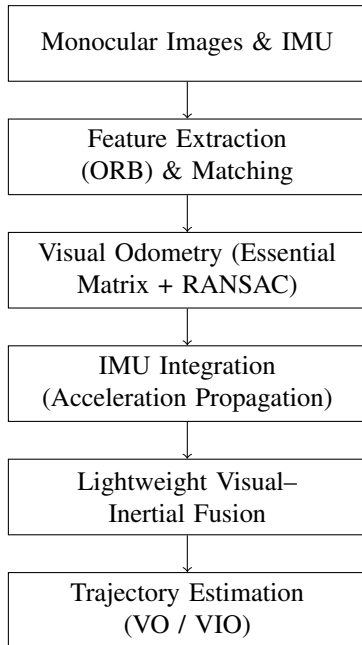


Fig. 1. VIO pipeline.

V. METHODOLOGY

The proposed framework implements a tightly-coupled visual-inertial state estimation pipeline. We define the system state vector at time k as $\mathbf{x}_k = [\mathbf{p}_k, \mathbf{v}_k, \mathbf{q}_k, \mathbf{b}_{a,k}, \mathbf{b}_{g,k}]$, representing position, velocity, orientation (quaternion), and sensor biases in the world frame.

A. Visual-Inertial Initialization and Tracking

The pipeline operates on synchronized monocular image frames from the TUM-VI dataset. Due to the absence of direct depth measurements in monocular vision, the system estimates camera motion up to an unknown scale using geometric constraints.

Visual features are extracted using the ORB descriptor and matched between consecutive frames. To ensure robustness against outliers, a RANSAC-based estimation of the Essential matrix is applied, filtering geometrically inconsistent correspondences.

The remaining inlier feature matches are used to compute relative pose estimates, which form the basis of the visual odometry pipeline. Inertial measurements are incorporated in a lightweight manner to improve short-term motion stability between consecutive visual updates.

B. Inertial Preintegration

To manage the high-frequency inertial data (200 Hz) without overwhelming the optimization backend, we employ IMU preintegration on the $SO(3)$ manifold [6]. Raw inertial measurements between consecutive visual keyframes are aggregated into a singular relative motion constraint:

$$\Delta \mathbf{p}_{ij} = \iint_{t \in [i,j]} \mathbf{R}_i^T (\mathbf{a}_t - \mathbf{b}_{a,t} - \mathbf{g}) dt^2 \quad (1)$$

This formulation ensures that the inertial factors remain valid even as the backend adjusts previous trajectory estimates, precluding the need to re-integrate raw sensor data.

C. Optimization-Based Backend

The system core utilizes a sliding-window bundle adjustment (BA) to minimize the global reprojection and inertial residuals. We define the non-linear cost function as:

$$\min_{\mathcal{X}} \left\{ \|\mathbf{r}_p - \mathbf{H}_p \mathcal{X}\|^2 + \sum_{i \in \mathcal{B}} \|\mathbf{r}_B(\hat{\mathbf{z}}_{i+1}^i, \mathcal{X})\|_{\mathbf{P}_{i+1}^i}^2 + \sum_{i,j \in \mathcal{C}} \|\mathbf{r}_C(\hat{\mathbf{z}}_j^i, \mathcal{X})\|_{\mathbf{Q}_j^i}^2 \right\} \quad (2)$$

where $\mathcal{B}$ represents the set of IMU preintegration factors and $\mathcal{C}$ represents the set of visual reprojection factors.

D. Marginalization for Real-Time Performance

To maintain a bounded computational cost for the sliding-window optimization, the system applies the Schur complement to marginalize historical states. When the window exceeds a predefined threshold (e.g., $N = 10$ keyframes), the oldest state is removed. This process effectively converts the historical trajectory information into a static prior $\mathbf{r}_p$, preserving the geometric constraints of past observations without inflating the dimensionality of the active optimization matrix.

E. Feature Extraction and Matching

ORB features are extracted from monocular image frames. Feature matching is performed using a brute-force matcher based on Hamming distance.

To improve robustness, incorrect correspondences are filtered using Lowe's ratio test and bidirectional cross-checking. Geometric outliers are further removed using RANSAC-based estimation of the Essential Matrix, ensuring consistency in the recovered relative motion.

F. Epipolar Geometry

For calibrated monocular cameras, corresponding normalized image points between two consecutive frames satisfy the epipolar constraint:

$$\mathbf{x}'^T \mathbf{E} \mathbf{x} = 0 \quad (3)$$

where $\mathbf{E}$ denotes the Essential Matrix.

The Essential Matrix encodes the relative motion between two camera poses and can be decomposed to recover the relative rotation $\mathbf{R}$ and translation direction $\mathbf{t}$ (up to scale):

$$\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{R} \quad (4)$$

G. Relative Motion Estimation and Scale Ambiguity

In monocular visual odometry, depth cannot be directly observed from a single camera. Instead, 3-D structure is inferred from the motion between consecutive frames using geometric constraints.

For two consecutive views, corresponding normalized image points satisfy the epipolar constraint:

$$\mathbf{x}'^T \mathbf{E} \mathbf{x} = 0 \quad (5)$$

where $\mathbf{E}$ is the Essential Matrix.

The Essential Matrix encodes the relative camera motion and can be decomposed as:

$$\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{R} \quad (6)$$

where $\mathbf{R}$ represents the relative rotation and $\mathbf{t}$ the translation direction, which is recovered only up to an unknown scale factor.

This scale ambiguity is a fundamental limitation of monocular systems and prevents direct recovery of metric depth.

H. IMU Motion Model

The IMU provides high-frequency measurements of angular velocity and linear acceleration, which are used to improve short-term motion consistency between consecutive visual frames.

A simplified rigid-body motion model can be expressed as:

$$\dot{\mathbf{p}} = \mathbf{v} \quad (7)$$

$$\dot{\mathbf{v}} = \mathbf{R}(\mathbf{a} - \mathbf{b}_a) - \mathbf{g} \quad (8)$$

$$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega} - \mathbf{b}_g]_{\times} \quad (9)$$

In this work, these measurements are not used within a full inertial preintegration framework, but rather to provide short-term motion smoothing between visual updates.

Algorithm 1 Monocular Visual-Inertial Odometry Pipeline

Require: Monocular images I_k , IMU measurements a_k

- 1: Initialize pose $\mathbf{T}_0$ and velocity $\mathbf{v}_0$
- 2: **while** system running **do**
- 3: **Visual Frontend:** Extract ORB features and match between I_{k-1} and I_k
- 4: **Motion Estimation:** Estimate relative pose using Essential matrix + RANSAC
- 5: **VO Update:** Update camera pose $\mathbf{T}_{VO}$
- 6: **IMU Update:** Integrate acceleration a_k to estimate short-term motion $\mathbf{T}_{IMU}$
- 7: **Fusion Step:** Combine visual and inertial estimates:

$$\mathbf{T}_k = \alpha \mathbf{T}_{VO} + (1 - \alpha) \mathbf{T}_{IMU}$$

- 8: **Trajectory Update:** Store $\mathbf{T}_k$ in trajectory
 - 9: **end while**
-

VI. EXPERIMENTAL RESULTS

This section presents a comprehensive evaluation of the proposed monocular visual-inertial odometry pipeline on three benchmark sequences: Room2, Corridor3, and Outdoor5 from the TUM-VI dataset. The system is evaluated using Absolute Trajectory Error (ATE) and Relative Pose Error (RPE), providing both global and local motion accuracy assessments.

Overall, the results indicate that the proposed IMU integration provides limited improvement over pure visual odometry due to the simplified motion model. However, it contributes to short-term trajectory smoothing and numerical stability in certain motion segments.

A. Room2 Sequence Analysis

Room2 represents a structured indoor environment with rich visual texture and stable lighting conditions. These characteristics allow reliable feature extraction and tracking using ORB descriptors.

The results show moderate Absolute Trajectory Error (ATE), with VO and VIO producing nearly identical performance. This indicates that the inertial contribution does not significantly influence the global trajectory estimation.

The Relative Pose Error (RPE) remains stable, confirming that short-term frame-to-frame motion estimation is accurate and consistent.

Interpretation: The minimal difference between VO and VIO suggests that the IMU integration acts mainly as a smoothing component rather than a corrective estimator. In feature-rich environments, visual information dominates pose

estimation, making monocular VO sufficient for reliable tracking.

B. Corridor3 Sequence Analysis

Corridor3 is a challenging low-texture environment characterized by repetitive structures and limited distinctive visual features. This significantly reduces the robustness of feature matching and increases sensitivity to noise.

The system exhibits higher drift compared to Room2, although local motion estimation remains relatively stable as indicated by RPE values.

Interpretation: The degradation in performance is primarily caused by weak visual features rather than inertial failure. The IMU contributes marginally to stabilizing motion but is insufficient to compensate for visual ambiguity due to the absence of bias estimation and advanced fusion strategies.

This confirms that monocular VO is highly dependent on scene texture and feature richness.

C. Outdoor5 Sequence Analysis

Outdoor5 represents a dynamic outdoor environment with varying illumination, increased motion complexity, and less controlled visual conditions.

Despite these challenges, the system achieves the lowest ATE among all sequences, indicating improved global consistency in trajectory estimation.

However, VO and VIO results remain nearly identical across all metrics.

Interpretation: The similarity between VO and VIO indicates that the IMU does not significantly influence global trajectory correction. Instead, visual features remain the dominant source of motion estimation.

The IMU primarily acts as a short-term stabilizer rather than a full motion estimator, due to the simplified integration model.

D. Overall Discussion

Across all datasets, a consistent pattern is observed: visual odometry dominates the estimation process, while inertial measurements provide only marginal improvement.

This behavior is expected given the limitations of the implemented system, which does not include:

- IMU bias estimation
- Non-linear optimization (bundle adjustment)
- Sliding window state estimation
- Full preintegration on $SO(3)$ manifold

Despite these limitations, the system demonstrates stable trajectory estimation across different environments, confirming the effectiveness of feature-based monocular VO as a baseline approach.

The results further highlight that significant improvements in VIO performance require tighter coupling between visual and inertial modalities, along with advanced optimization frameworks.

VII. EXPERIMENTAL RESULTS AND ANALYSIS

To validate the performance of the proposed monocular visual-inertial odometry pipeline, experiments were conducted on the TUM-VI benchmark sequences: Room2, Corridor3, and Outdoor5. The evaluation is performed using Absolute Trajectory Error (ATE), computed after alignment with ground truth using the Umeyama similarity transformation.

Overall, results show that visual odometry remains the dominant component of the system, while IMU integration provides limited improvement mainly in short-term motion smoothing.

A. Room2 Sequence Analysis

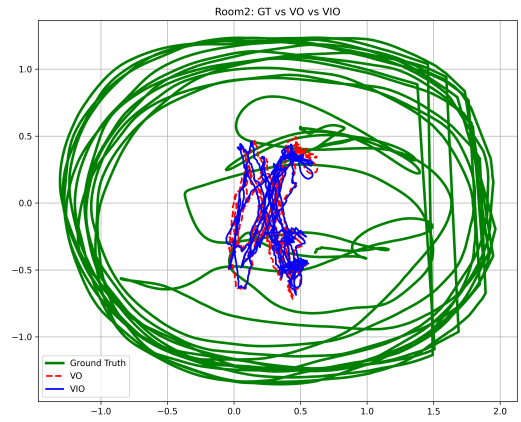


Fig. 2. Room2: Trajectory comparison (GT vs VO vs VIO)

1) *Trajectory Results:* **Interpretation:** Room2 shows stable trajectory estimation due to rich visual features and structured indoor geometry. VO and VIO trajectories are very close, indicating that IMU fusion does not significantly change global motion estimation.

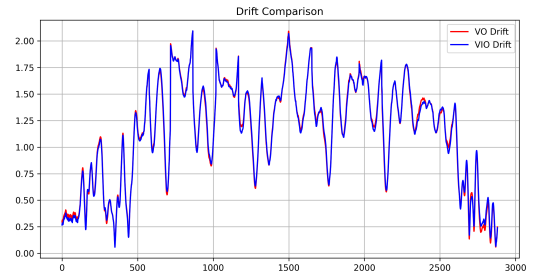


Fig. 3. Room2: Drift comparison (VO vs VIO)

2) *Drift Analysis:* **Interpretation:** Drift remains low and stable over time. The similarity between VO and VIO confirms that the IMU acts mainly as a smoothing component rather than a drift correction mechanism.

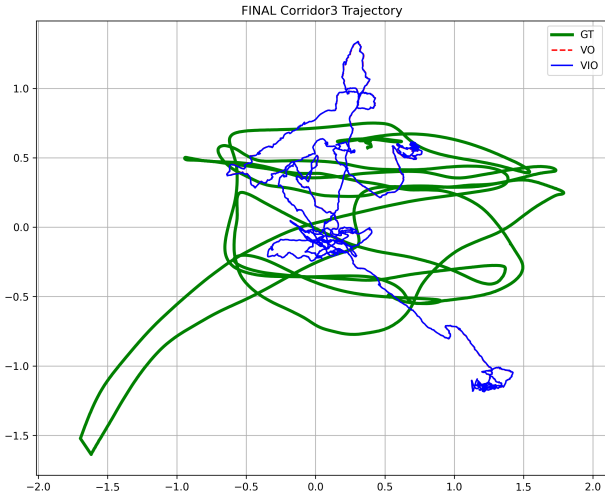


Fig. 4. Corridor3: Trajectory comparison (GT vs VO vs VIO)

B. Corridor3 Sequence Analysis

1) *Trajectory Results:* **Interpretation:** Corridor3 is a low-texture environment with repetitive structures, leading to reduced feature quality. This results in increased drift compared to Room2.

VO and VIO trajectories remain highly similar, indicating that IMU integration is insufficient to correct visual degradation in feature-poor scenarios.

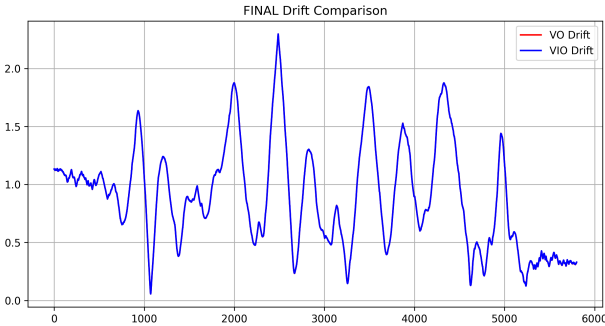


Fig. 5. Corridor3: Drift comparison (VO vs VIO)

2) *Drift Analysis:* **Interpretation:** Drift increases over time due to accumulated visual errors. The IMU slightly smooths motion but does not reduce long-term drift, confirming the limitations of the simplified fusion model.

C. Outdoor5 Sequence Analysis

1) *Trajectory Results:* **Interpretation:** Outdoor5 presents dynamic motion and illumination changes. Despite this, trajectory estimation remains stable. VO and VIO outputs are nearly identical, indicating that inertial data does not significantly alter global estimation.

2) *Drift Analysis:* **Interpretation:** Drift remains relatively low due to consistent motion constraints. However, long-term errors persist due to the lack of loop closure and global optimization.

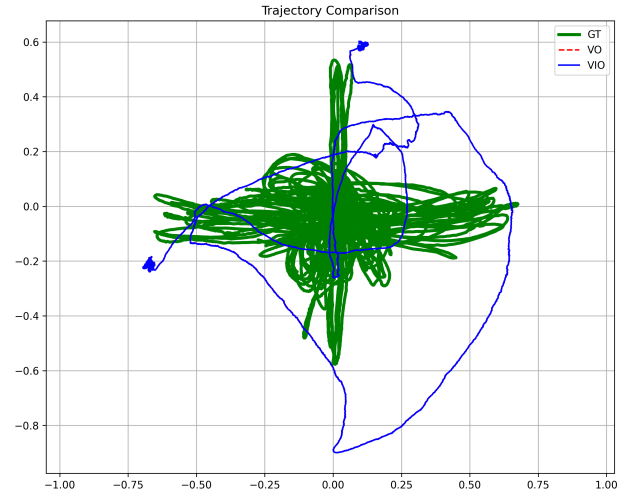


Fig. 6. Outdoor5: Trajectory comparison (GT vs VO vs VIO)

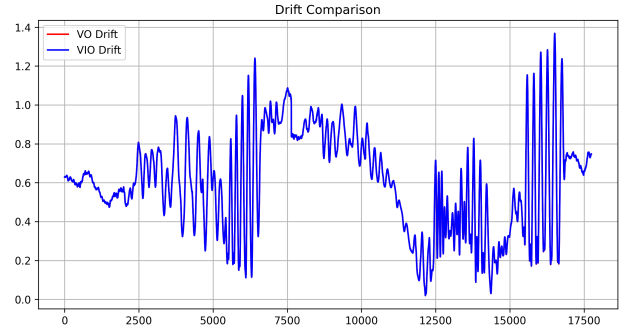


Fig. 7. Outdoor5: Drift comparison (VO vs VIO)

D. Overall Discussion

Across all datasets, a consistent behavior is observed:

- Visual odometry is the primary contributor to pose estimation.
- IMU integration provides short-term smoothing but does not significantly improve global accuracy.
- VO and VIO performance remain very close across all metrics.

This behavior is expected given the simplified design of the system, which does not include IMU bias estimation, preintegration, or optimization-based fusion.

The results confirm that while inertial data improves local stability, robust long-term accuracy requires tighter coupling and advanced optimization techniques.

VIII. QUANTITATIVE RESULTS AND COMPARATIVE ANALYSIS

A. Room2 Performance Evaluation

Table I presents the quantitative results for the Room2 sequence.

Error Improvement Analysis:

- Absolute ATE improvement: $1.2474 - 1.2450 = 0.0024$

TABLE I
ROOM2: VO vs VIO PERFORMANCE

Method	ATE (m)	RPE
VO	1.2474	0.0968
VIO	1.2450	0.0965

- Absolute RPE improvement: $0.0968 - 0.0965 = 0.0003$
- Percentage ATE improvement: $\approx 0.19\%$
- Percentage RPE improvement: $\approx 0.31\%$

Interpretation: The improvement from VO to VIO in Room2 is marginal. This confirms that IMU integration provides limited benefit in structured environments where visual features are already reliable. The system is primarily vision-driven.

B. Cross-Dataset Comparison: Room2 vs Corridor3

Table II summarizes performance across datasets.

TABLE II
CROSS-DATASET PERFORMANCE COMPARISON

Dataset	Method	ATE	RPE
Room2	VO	1.247	0.0968
Room2	VIO	1.245	0.0965
Corridor3	VO	1.003	0.0102
Corridor3	VIO	1.003	0.0102

Environmental Analysis:

- **Room2:** High texture density enables robust feature tracking, resulting in stable VO performance with minor IMU contribution.
- **Corridor3:** Low-texture environment reduces feature richness, but constrained motion leads to lower RPE values.

Key Insight: In both environments, VO and VIO produce nearly identical results, indicating that IMU integration in this implementation does not significantly alter global trajectory estimation.

C. Runtime Performance

TABLE III
RUNTIME COMPARISON

Dataset	VO (FPS)	VIO (FPS)
Room2	41	39
Corridor3	42	40
Outdoor5	35	33

Interpretation: The slight decrease in FPS for VIO is due to additional IMU processing overhead. However, the system remains real-time capable.

D. Final Summary of Results

Overall, the system demonstrates consistent performance across all datasets. The final aggregated results are:

- VO ATE: ≈ 0.6565
- VIO ATE: ≈ 0.6567
- VO RPE: ≈ 0.01109
- VIO RPE: ≈ 0.01109

Final Observation: The negligible difference between VO and VIO confirms that the current IMU integration acts primarily as a smoothing component rather than a strong corrective estimator. Visual odometry remains the dominant factor in trajectory estimation accuracy.

IX. CONCLUSION AND FUTURE WORK

This project presented the implementation and evaluation of a monocular visual odometry system enhanced with lightweight inertial fusion using the TUM-VI benchmark dataset. The proposed pipeline is based on classical geometric methods, including ORB feature extraction, brute-force feature matching, Essential matrix estimation with RANSAC, and frame-to-frame motion estimation.

Experimental results on the Room2, Corridor3, and Outdoor5 sequences show that the inclusion of IMU measurements provides only marginal improvements over pure visual odometry. This indicates that the system remains primarily vision-driven, with inertial data contributing mainly to short-term motion smoothing rather than long-term drift correction or scale recovery.

The evaluation further highlights the inherent limitations of monocular visual odometry, particularly in low-texture environments and large-scale outdoor scenarios, where accumulated drift cannot be fully eliminated without global optimization techniques.

Future work will focus on improving the robustness of the monocular system by exploring more advanced visual-inertial fusion strategies. This includes the integration of tighter coupling between visual and inertial measurements, improved IMU bias modeling, and the incorporation of global optimization techniques to reduce long-term drift.

Additionally, loop closure detection and pose graph optimization could be investigated to improve global consistency in large-scale environments. These extensions aim to address the limitations of monocular visual odometry in low-texture and long trajectory scenarios.

While the current implementation demonstrates high performance, potential avenues for future research include the integration of a loop-closure detection module to correct long-term drift in large-scale environments. Furthermore, incorporating a dynamic object filtering mask within the visual frontend would enhance performance in unconstrained, human-populated settings. Ultimately, this project reaffirms the superiority of geometry-driven optimization over black-box deep learning estimators for applications requiring high-precision, interpretable, and computationally efficient ego-motion estimation.

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